Task 3: Communication Data, Signals, Events and Protocols

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This document lists the data, signals and events the leader and follower truck need to communicate in each scenario. For each one I have written down what it is, who sends it, who receives it and the timing it has to meet. I also wrote out the protocol for each scenario. All three scenarios were then modelled as timed automata in UPPAAL and verified.

Scenario 1: Normal Platooning

In normal platooning the leader keeps sending its speed and position while the follower stays in its lane using the camera and keeps a safe distance. Everything is working, nothing has failed.

Name	**Type**	**Sender**	**Receiver**	**Meaning**	**Timing constraint**
leader_speed_position	Data	Leader Truck	Follower Truck	The leader keeps sending its current speed and position so the follower can match it.	Sent and received within 100ms (TR3)
camera_frame	Data	Camera	Lane Detection Model	Image frame taken at 30 FPS for lane detection.	Captured and processed within 33ms (TR1)
lane_position	Data	Lane Detection Model	Follower Truck	The detected lane markings used for steering.	Within one frame cycle, 33ms (TR2)

steering_correction	Signal	Follower Truck	Steering Controller	Command to keep the truck inside its lane.	Applied within 33ms (TR2)
following_distance	Data	Dist/Speed Controller	Follower Truck	The gap to the leader, checked against the 5m minimum.	Continuous (FR3)
distance_below_min	Event	Dist/Speed Controller	Follower Truck	Fires when the gap drops below 5m so the truck reduces speed.	Immediate (SR1)
lane_lost_alert	Event	Lane Detection Model	Leader Truck	Raised when lane markings are missing for more than 3 frames, so the truck slows and alerts.	After 3 frames (SR2)
speed_command	Signal	Dist/Speed Controller	Throttle Actuator	Adjusts speed to match the leader without going faster than it.	Speed sync within 100ms (TR3, SR3)

Protocol
The leader sends its speed and position at least every 100ms (TR3). The follower runs a perception loop at 30 FPS, so each frame is processed within 33ms (TR1) and a steering correction is applied in the same frame (TR2). The distance and speed controller keeps checking the following distance. If it drops below 5m the truck reduces speed automatically (SR1), otherwise it matches the leader's speed and never goes

faster than it (FR4, SR3). If the lane markings are lost for more than 3 frames the truck raises an alert and slows down (SR2).

Scenario 2: Communication Failure

Here the wireless link drops and the follower stops getting beacons from the leader. It has to notice this within 500ms and switch to a fallback mode on its own, slowing down safely until either the link comes back or it stops.

Name	Type	Sender	Receiver	Meaning	Timing constraint
leader_speed_position	Data	Leader Truck	Follower Truck	The normal speed and position beacon. When it stops arriving, that means the link is lost.	Period 100ms or less
timeout_reset	Event	Follower Truck	Timeout Timer	Resets the timeout timer every time a beacon comes in (FR1).	On every received signal
signal_timeout	Event	Timeout Timer	Follower Truck	Fires when no beacon has arrived for 500ms, which counts as a communication failure.	Within 500ms of last signal (TR1, FR2)
switch_to_fallback	Signal	Follower Truck	Dist/Speed Controller	Tells the truck to leave platooning mode and go into fallback mode.	Within 100ms of detection (TR2)
decelerate_command	Signal	Follower Truck	Throttle Actuator	Slows the truck down smoothly toward a safe speed or a	Starts within 600ms of last signal (TR3, SR2)

				stop, no hard braking.	
camera_lane_position	Data	Camera / Lane Detection Model	Follower Truck	Lane keeping keeps running off the camera during fallback.	Frame processing 33ms or less (FR5, SR3)
link_restored	Event	Wireless Link	Follower Truck	The beacon comes back before 10s and the truck goes back to platooning.	Before 10s
full_stop_command	Signal	Follower Truck	Throttle Actuator	If the link is still gone after 10s, bring the truck to a full stop.	At 10s (SR4)

Protocol
The follower resets its timeout timer every time a beacon comes in (FR1). If nothing arrives for 500ms the timeout fires and this counts as a communication failure (TR1, FR2). The follower switches to fallback mode within 100ms (TR2) and starts slowing down smoothly within 600ms of the last signal (TR3, SR1, SR2). The camera keeps doing lane keeping the whole time (FR5, SR3). If the link comes back before 10s the truck goes back to platooning, and if it does not the truck comes to a full stop at 10s (SR4).

Scenario 3: Emergency Braking

An obstacle suddenly appears ahead of the leader. The leader brakes hard and sends a brake signal to the follower. The follower has to brake within 200ms, and if it misses the signal it still has to spot the leader braking through its camera and brake on its own.

Name	Type	Sender	Receiver	Meaning	Timing constraint
emergency_brake_signal	Signal	Leader Truck	Follower Truck	A high priority message sent the moment the leader brakes,	Received and acted on within 200ms (TR1)

				telling the follower to brake too.	
obstacle_detected	Event	Leader Truck	Leader Truck	Fires when an obstacle suddenly shows up ahead of the leader.	Immediate (FR1)
brake_signal_received	Event	Wireless Link	Follower Truck	The follower gets the brake signal in time (Path A).	Within 200ms (TR1)
brake_signal_missed	Event	Follower Truck	Follower Truck	No brake signal arrived in time, so the truck falls back to the camera (Path B).	After signal timeout
camera_deceleration_detected	Event	Camera / Lane Detection Model	Follower Truck	The camera sees the leader slowing down, for example its brake lights or the gap closing.	Frame 33ms or less (TR3), braking within 300ms (TR2)
precautionary_stop	Event	Follower Truck	Braking System	Both paths failed within	If no braking by 400ms (SR2)

				400ms, so the truck does a safe controlled stop (Path C).	
apply_braking_command	Signal	Follower Truck	Braking System	Applies controlled braking force based on how hard the leader is braking.	Smooth and controlled (FR5, SR3)

Protocol
There are two backup paths so that the wireless link and the camera are not single points of failure (NFR2, SR1). In Path A the leader sends the brake signal when it brakes, and if the follower gets it, it brakes within 200ms (TR1). In Path B, if the signal is missed, the camera spots the deceleration (frame within 33ms, TR3) and braking is triggered within 300ms (TR2). In Path C, if neither path brakes within 400ms, the truck does a precautionary controlled stop (SR2). In all cases the braking is smooth and proportional and the truck stays in its lane (FR5, SR3, SR4).

Formal Verification in UPPAAL

Each scenario was built as a network of timed automata in UPPAAL and checked. For all three I confirmed there is no deadlock, that the important safe states can be reached (fallback mode, safe stop, speed matching) and that the timing bounds hold, for example failure detection within 500ms, the mode switch within 100ms, signal based braking within 200ms, camera based braking within 300 to 400ms and frame processing within 33ms.
Model files: scenario1_normal_platooning.xml, scenario2_communication_failure.xml, scenario3_emergency_braking.xml